

Internship report

Bluetooth Low Energy Channel Sounding for Drone
Ranging Optimisation

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Table of Acronyms

Acronym	Full term
ACL	Asynchronous Connection-oriented Link
AoA	Angle of Arrival
AoD	Angle of Departure
BLE	Bluetooth Low Energy
CS	Channel Sounding
DF	Direction Finding
FFO/CFO	(Fractional/Carrier) Frequency Offset
GATT	Generic Attribute Profile
GNSS	Global Navigation Satellite System
IFFT	Inverse Fast Fourier Transform
IMU	Inertial Measurement Unit
IQ	In-phase/Quadrature
NCS	nRF Connect SDK
PBR	Phase-Based Ranging
PCT	Phase Correction Term
PDOP	Position Dilution of Precision
RSSI	Received Signal Strength Indicator
RTK	Real-Time Kinematics
RTOS	Real-Time Operating System
RTT	Round-Trip Time
SNR	Signal-to-Noise Ratio
UART	Universal Asynchronous Receiver-Transmitter
UAV	Unmanned Aerial Vehicle
UWB	Ultra-wideband

1. Introduction

NTNU was established in 1996 through the merger of six institutions, including the Norwegian Institute of Technology (NTH, founded in 1910), and today ranks among Norway's leading research universities. Reflecting Norway's strong maritime and offshore industry, the university has developed recognized expertise in marine and ocean technology. This orientation is closely tied to the objectives of the UAV-Lab, whose work targets multiple themes but one of his main subject is the autonomous navigation in operational environments where conventional positioning fails, such as flight over water or maritime approach scenarios, where visual methods struggle and GNSS may be unavailable or jammed. It is within this context that the PARNAV project seeks a resilient, radio-based alternative for localization in GNSS-denied conditions.

The PARNAV project stems from the will to reduce our dependence on GNSS when navigating. Its goal is to use phased-array radio systems for resilient localization and navigation of autonomous systems in GNSS-denied environments.

GNSS is by far the most widespread way to obtain an absolute position. Standard receivers reach an accuracy on the order of a few meters, the infrastructure is already deployed globally, and the cost of integrating a receiver is negligible. This raises a legitimate question: why look for an alternative at all?

The answer lies in the vulnerabilities of the system. GNSS signals reach the receiver with very low power, which makes them easy to disrupt. Two threats are particularly relevant. Jamming consists of emitting a strong radio signal on the GNSS frequencies to drown out the satellite signals, so the receiver can no longer compute a position. Spoofing is more insidious: a counterfeit signal mimics the real satellite signals to deceive the receiver into computing a false position, without it necessarily detecting the attack. To these deliberate threats one can add the simple case of indoor or obstructed environments, where the signal is blocked or heavily degraded and GNSS becomes unusable regardless of any malicious intent.

A further limitation concerns precision. Reaching centimeter-level accuracy requires Real-Time Kinematics (RTK), which relies on a nearby fixed base station transmitting corrections to cancel the drift affecting the raw GNSS solution. This dependency on local infrastructure restricts where high-accuracy positioning can actually be achieved.

This is where the idea behind BLEDAR takes shape: exploiting low-cost, mass-produced Bluetooth Low Energy devices to provide positioning without relying on satellites. The goal is to fuse two BLE-based measurements, the distance obtained through Channel Sounding and the direction obtained through Direction Finding, with inertial sensing, so that a single ground-based unit can localize an autonomous vehicle in three dimensions even when GNSS is denied.

The goal of my internship is to develop a BLE system that leverages the Channel Sounding feature to estimate the distance between fixed beacons, referred to as reflectors, and a mobile unit, referred to as the initiator. The hardware platform is the Nordic Semiconductor nRF54L15DK development kit, running the Zephyr real-time operating system. The embedded firmware is written in C and is responsible solely for executing the Channel Sounding procedures and streaming the raw measurement data over UART. The distance estimation itself is performed off-board by a Raspberry Pi running Python, which reads this stream, monitors the system in real time, and computes the range estimates from the raw CS data. This overall architecture is illustrated in Figure 1, where the mobile initiator, carried by the drone together with the Raspberry Pi, ranges against several fixed reflectors deployed on the ground.

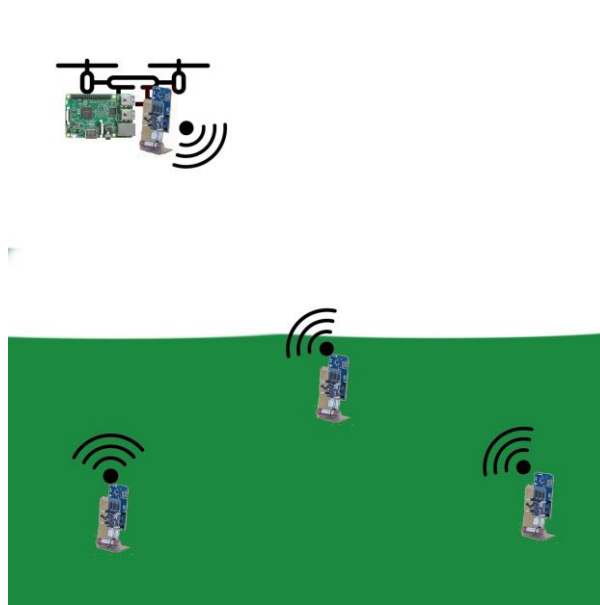


Figure 1: Overview of the ranging system. The airborne unit carries the initiator (nRF54L15DK) connected to a Raspberry Pi over UART, while several ground-based reflectors (nRF54L15DK) act as fixed beacons. Each initiator-reflector link performs a Channel Sounding exchange from which a distance is estimated.

This document relates the process followed to obtain accurate and fast ranging measurements through seven main stages, starting with the state of the art of positioning methods, then explaining what Channel Sounding is and its place in the Bluetooth specification. We then define the objectives and scope of this work, before detailing how the Channel Sounding procedure was implemented and optimized, and how the raw IQ data is processed to extract a distance estimate. Finally, we present our experimental setup, the tests performed and their results, discuss the limitations of the study, and conclude on what this work achieved.

2. State of the art of positioning methods

Estimating the position of a system relies on a two-step logic: we first measure a physical quantity, then convert these measurements into a position estimate. This section reviews these measurement principles and the geometric constructions they enable, before describing the operational systems built upon them.

2.1 Measurement principles

2.1.1 Inertial Measurement Unit (IMU)

An IMU is a component designed to measure acceleration (accelerometer) and angular velocity (gyroscope). These measurements are internal to the system and require no external reference, which makes them always available and impossible to jam.

2.1.2 Propagation time

The distance between two radio units can be obtained from the time taken by a signal to travel between them, since it propagates at a known speed, namely the speed of light in the medium. Measuring a one-way time of flight requires the transmitter and the receiver to share a common time reference, since the departure is timestamped by one device and the arrival by the other. Any offset between the two clocks translates directly into a distance error, and at the speed of light a discrepancy of one nanosecond already corresponds to about 30 cm.

Round-trip timing (RTT) avoids this constraint. The initiator starts its timer when the signal is transmitted, the reflector replies after a known and fixed turnaround delay, and the initiator stops its timer upon reception. Both timestamps are produced by the same clock, so any offset between the two devices cancels out. The time of flight is then obtained by subtracting the turnaround delay from the measured round-trip duration and dividing the result by two. The synchronisation requirement is therefore traded for the need to know the reflector's turnaround delay precisely.

2.1.3 Phase difference across antennas

When the same signal is received by several antennas placed at known positions, the wavefront does not reach them all at the same instant. The extra path travelled to reach one element rather than another depends on the direction from which the wave arrives, and since a path difference translates into a phase shift, measuring the phase difference between the elements gives access to that direction. Using several elements arranged in a planar array allows both azimuth and elevation to be estimated. This is the measurement underlying direction finding, which yields an angle rather than a distance.

2.1.4 Phase across frequencies

The phase of a received signal also carries distance information. A signal that has travelled a given distance arrives with a phase shift proportional to that distance, but this shift alone is not exploitable: a phase is only known modulo one full rotation, so a great many distances produce the same measured value.

The way out is to repeat the measurement at several frequencies. Since the phase shift depends on both the distance and the frequency, comparing the phases obtained across a set of frequencies isolates the distance. The rate at which the phase rotates from one frequency to the next is proportional to the propagation distance, which turns a set of ambiguous phase measurements into a single distance estimate. The ambiguity is not removed entirely, since the spacing between frequencies sets a maximum distance beyond

which the pattern repeats itself. This principle is known as phase-based ranging and constitutes the core of the work presented in this report. This principle is more detailed in section 3: Channel Sounding.

2.2 From measurements to a position

2.2.1 Dead reckoning

Inertial measurements do not give a position directly. Integrating the angular velocity yields the attitude of the system, and integrating the acceleration twice yields first its velocity, then its position, provided the initial state is known.

This construction is self-contained, but its error is unbounded. Sensor biases and noise are integrated along with the useful signal, so the estimated position slowly drifts away from the true one, and the drift grows with time. An inertial solution is therefore excellent over short intervals but has to be corrected periodically by an absolute measurement, which is exactly the role played by the radio-based methods described below.

2.2.2 Multilateration

When the distances to several reference points of known position are available, each measurement constrains the system to a sphere centred on the corresponding anchor. The position is found where these spheres intersect. In practice, at least three anchors are needed to resolve a three-dimensional position, and their spatial distribution matters as much as the quality of the individual measurements: anchors that are too close together, or aligned, yield an intersection that is very sensitive to measurement noise.

2.2.3 Triangulation

Triangulation originally consisted in measuring the bearing, that is the angle relative to north, of several known landmarks. Each bearing traces a line of position on the map, and the observer's location is found where these lines intersect. The same principle applies to radio positioning: each angular measurement constrains the target to a direction, and the position is obtained by intersecting these directions.

A single station providing both azimuth and elevation defines a line in space, which becomes a full three-dimensional position as soon as one distance along that line is known. This observation is central to the approach followed in this work, since it means that one station combining an angle and a range can replace a network of anchors.

2.3 Operational systems

2.3.1 GNSS and RTK-GNSS

GNSS applies multilateration on a global scale, using satellites as anchors and the propagation time of their signals as the measured quantity. Its limitations have already been discussed in the introduction. RTK-GNSS refines this solution to the centimetre level by exploiting the carrier phase and by applying corrections broadcast by a nearby base station, at the cost of a dependency on local infrastructure. In this work, RTK-GNSS serves as the ground truth reference against which the alternative methods are evaluated, since its accuracy is roughly an order of magnitude better than theirs.

2.3.2 Ultra-wideband

Ultra-wideband (UWB) is the established reference for local radio ranging. It applies multilateration using ground anchors and reaches decimeter-level accuracy on commercially available hardware. Its limitation is the infrastructure it demands: several anchors must be deployed around the operating area and their positions accurately surveyed.

Moreover, the accuracy of the result depends heavily on the geometry of the anchors relative to the target. This sensitivity is commonly quantified by the position dilution of precision (PDOP), which relates the measurement noise to the resulting position error. For the five-anchor UWB setup used as a comparison in the PARNAV experiments, the PDOP is 1.61 for a vehicle located inside the anchor network, whereas a single-station phased-array system reaches 22.00 at the same location. Geometry alone therefore accounts for more than an order of magnitude in position error, which explains why UWB requires anchors distributed all around the operating volume. This is a real operational constraint, for instance when a vehicle approaches a dock and no anchor can be placed on the water side.

2.3.3 Bluetooth Low Energy

Bluetooth Low Energy offers a much cheaper alternative, since it relies on mass-produced components that are already present in a wide range of devices. This technology is more detailed in section 3: Channel Sounding.

2.4 Distance estimation from phase measurements

The Channel Sounding specification standardizes how the phase measurements are acquired but leaves their conversion into a distance entirely to the implementer. Several families of estimators exist, among which two are considered in this work: a frequency-domain method, which applies an inverse Fourier transform to the measured channel response so that the propagation delay appears as a peak in the resulting impulse response, and a Bayesian estimator, which works directly on the raw tones. Choosing an estimator is one of the keys to proper ranging, since it determines how well physical aberrations such as multipaths can be handled. This topic is developed in section 6: Distance estimation.

3. Channel Sounding in the Bluetooth specification

3.1 The history of ranging in the Bluetooth specification

The Bluetooth Low Energy (BLE) has been originally designed as an efficient way to transfer data through multiple devices in terms of energy consumption and data exchange speeds. The ability to measure different distances was added through time while the specification added more and more features. This section follows the timeline of the specification. Starting from Received Signal Strength Indicator (RSSI) to Direction Finding (DF or AoA/AoD), which measures an angle but not a range, and finally to Channel Sounding (CS), which measures the range itself.

3.1.1 The Received Signal Strength Indicator

The first way to obtain a distance from a Bluetooth link was never designed for it. Every radio receiver can evaluate the power of the signal received. This power is expressed in dbm and represents the intensity of the radio link. Based on this observation people can estimate a distance between two antennas with lookup table representing the loss in terms of signal power received.

This way of estimating distance is easy and works for low precision needed. One of the big weaknesses of this system is his dependence of the surrounding environment. In fact, if the two bluetooth systems are placed with no direct clearance the estimation will be full of errors due to multipath issues destructive and constructive interferences. Even the density of the air can modify the distances perceived. All that noise makes this system really limited for distances measurement.

3.1.2 The direction finding, a measurement of the angle

The Bluetooth standard 5.1 published in 2019, introduced the first feature deliberately intended for localization. Direction finding is based on an idea which is that a wavefront will have a light phase difference between the antenna receiving it. This difference allows to reveal the direction of the emitter as explained in section 2.1.3: Phase Across Antenna.

The standard exposes raw phase data rather than a processed result, and Channel Sounding follows the same principle, which allows us to try and use different signal treatment to estimate distances like treating all the data off the Bluetooth board to process multiple data fusion from IMU like example to get more accurate results.

3.1.3 The Channel Sounding, the measurement of the range itself

The Bluetooth standard 6.0, published in 2024, finally adds the quantity that was missing. Channel Sounding is a bidirectional exchange between two devices, from which a distance is computed. It does not invent a new physical principle. It combines the two that were already presented in section 2.1: Measurement principles: the round-trip time, which is coarse but never ambiguous, and the phase measured across several frequencies, which is precise but repeats itself beyond a certain distance. Each one covers the weakness of the other. The specification interleaves them inside a single procedure, and the way this interleaving is built is the subject of section 3.2: Anatomy of a CS procedure.

Three properties of this feature matter for the present work. The first one is interoperability: two devices from different manufacturers can range against each other if both are compatible with Bluetooth 6.0 or higher, which makes the argument about cheap mass-produced components a real thing and not only a promise. The second one is security: the ranging sequences are randomized, so an attacker cannot replay or amplify the exchange to make a device appear closer than it really is. The third one is the most important

thing for us. The specification delivers the raw phase measurements, and the conversion is left to the application. We will talk about the conversion algorithms in section 6: Distance estimation.

Seen from the end, the three additions are not a list of features but a progression. The RSSI gave a proximity, the Direction Finding gave an angle, the Channel Sounding gives a range. Section 2.2.3 showed that a single station providing an azimuth, an elevation and one distance is enough to define a complete three-dimensional position. This is exactly the architecture of the BLEDAR system, and this is why the ranging part developed during this internship is built on Bluetooth and not on another radio technology. The specification, however, only describes how the measurements are acquired and that exactly what we will take look into the following section to understand the CS procedures.

3.2 Anatomy of a CS procedure

The three important things to understand in the CS procedure are:

- The Round-Trip Time (RTT) and why this algorithm is useful for the CS procedure
- How it's construct, what are the subprocess, steps and what really constitute a procedure
- The CS parameters and how each one impacts the others.

And we already talked about RTT in section 2.1.2. We will take time to decompose how a procedure is built in the following sections and develop how the CS parameters impact each other in section 5: Implementation and optimisation of the Channel Sounding procedure.

3.2.1 How is a CS procedure constructed ?

Channel Sounding is organised as a hierarchy of four nested time units: a procedure contains one or more events, each event is split into subevents, and each subevent is divided into steps. The step is the smallest unit : it is where the actual radio exchange between the initiator and the reflector takes place. Every start time in this hierarchy (procedure, event, subevent and step) is anchored to a connection event of the ACL link between the two devices, so the whole CS activity rides on top of the existing BLE connection.

The step and its mode. A step is a single bilateral exchange: the initiator transmits first, the reflector responds. What is exchanged depends on the step's mode :

- Mode-0: calibration step. It measures the frequency offset (FFO/CFO) between the two clocks so the following steps can be corrected.
- Mode-1: RTT step. It exchanges CS_SYNC packets to measure the round-trip time.
- Mode-2: Tone Step. It exchanges CS tones to measure the phase (PBR).
- Mode-3: A combined step. It exchanges a CS_SYNC and a tone together (RTT + PBR in one step).

3.2.2 How is a subevent built?

A subevent always starts with one to three mode-0 steps for calibration, then runs a repeating pattern of a main-mode step type interleaved with a sub-mode step type. The number of main-mode steps before a sub-mode step is inserted is set by a parameter. A subevent holds between 2 and 160 steps. The calibration is redone in every subevent: mode-0 always reappears at the start of the next one.

3.2.3 What constitutes a procedure?

Each step uses a different channel of the spectrum (the CS band has 72 usable channels). A single subevent only samples a few points of the spectrum. It represents the analysis of one part of it. To estimate a distance by phase, the system needs the phase measured across a wide range of frequencies, so the subevents are repeated, sweeping the channel map until enough of the spectrum has been covered. A procedure is this complete set of subevents, up to 256 steps, that together produce one distance estimate. The procedure is then restarted at a regular interval to produce the next estimate.

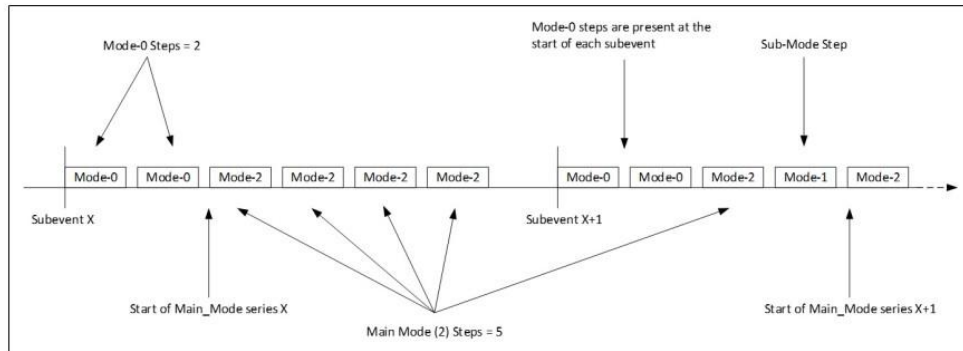


Figure 2: CS mode sequencing example with two mode-0 steps, five Main_Mode steps (of type mode-2), and one Sub_Mode step (of type mode-1)

4. Objectives and scope

The previous sections showed that many methods are available to obtain positioning data, but that each of them carries a weakness which becomes critical as soon as the system has to be resilient: inertial navigation drifts without bound, GNSS can be jammed or spoofed, and UWB demands an infrastructure of surveyed anchors that is not always deployable.

This work builds instead of a standard consumer technology, Bluetooth Low Energy. Since the release of the Bluetooth Core Specification 6.0 we have the opportunity to use the new introduced feature which is the Channel Sounding procedure. This new feature allows to accurately measure the distance between two 6.0 bluetooth systems which in our cases will be some nRF54L15DK. Combining this technology with IMU we could manage to fusion all the data avoiding natural IMU drift by estimating the drone position with CS and AoA.

The main goal will be to develop embedded software allowing us to get great sampling/precision rate. The sample code provided by nordic runs on one pair of initiator/reciever running at 0.5Hz and we have to manage to get better performances from this key feature.

The section 3: Channel Sounding in the Bluetooth specification showed us the construction of the CS procedure. Now the goal will be to tune the system so that it can serve three distinct use cases: a UAV flying outdoors, a boat approaching a dock, and a drone operating indoors. These scenarios differ in terms of precision and sampling rated needed, so the specification of the Bluetooth will allow us to understand the key parameters to tune.

This leads to the question: how can a Channel Sounding procedure be configured to extract the best possible performance for our applications?

Two main difficulties follow

- The first is the ranging itself: phase measurements are ambiguous by nature and can be corrupted by noise and multipath.
- The second is the multiple beacons configuration: the controller imposes constraints on how the procedures run in parallel. The antenna time needs to be shared by all the beacons without corruption of the CS sequences.

The scope of this internship restricted to the ranging component. The system delivers, for each beacon, an IQ measurement characterized by the phase measurement, and those data will be processed to manage to get a distance estimation.

Within this scope, the work is evaluated against three main criteria: the accuracy of the individual range estimates, the update rate at which they are produced, and the number of beacons that can be handled concurrently.

5. Implementation and optimization of the Channel Sounding procedure

This section describes how the CS feature has been implemented. It explains how the configuration has been planned and justifies each setting against the operational constraints of the three use cases. It also covers how the system scales with the number of reflectors, explores the unwanted issues that appear when the CS procedure is enabled, and how we managed to extract the best performance from our system.

5.1 Skeleton of the embedded software

The Nordic development kit runs the nRF Connect SDK, whose application layer is built on the Zephyr RTOS. Zephyr provides a set of primitives (semaphores, message queues, work queues) whose blocking calls all accept an explicit timeout argument. This allows the execution of each software stage to be deterministic and bounded, which makes the latency of a measurement cycle predictable. Second, no code path can block indefinitely: every wait on an asynchronous event is expressed as a « `k_sem_take()` » or "`k_msgq_get()`" with a finite timeout, so a controller that fails to report a result degrades into a recoverable error rather than a stalled system.

The firmware is organized in four layers, each corresponding to a distinct responsibility:

- API layer. Direct calls to the Bluetooth host API exposed by the SDK (connection establishment, CS capability exchange or security...). These functions do not carry application logic.
- Procedure layer. Application-level operations that encapsulate one or several API calls, together with the parameter checking, retry policy, and completion of synchronization they require. A procedure is written as a sequential, blocking function so that the setup sequence reads in the order in which it occurs.
- Callback layer. Asynchronous notifications raised by the host and the controller (connection events, CS configuration completion...). These callbacks execute in a specific thread context, so they perform no processing beyond copying the payload into a queue and releasing the semaphore awaited by the corresponding procedure.
- Main orchestration. A state machine that sequences the procedures, arbitrates between the beacons of the network, and handles error recovery.

This separation isolates the two execution contexts of the application: the sequential context, in which the setup logic is written, and the asynchronous context, in which the controller reports its results. The interface between them is reduced to a small number of semaphores and message queues, which is what allows every wait to be given a timeout.

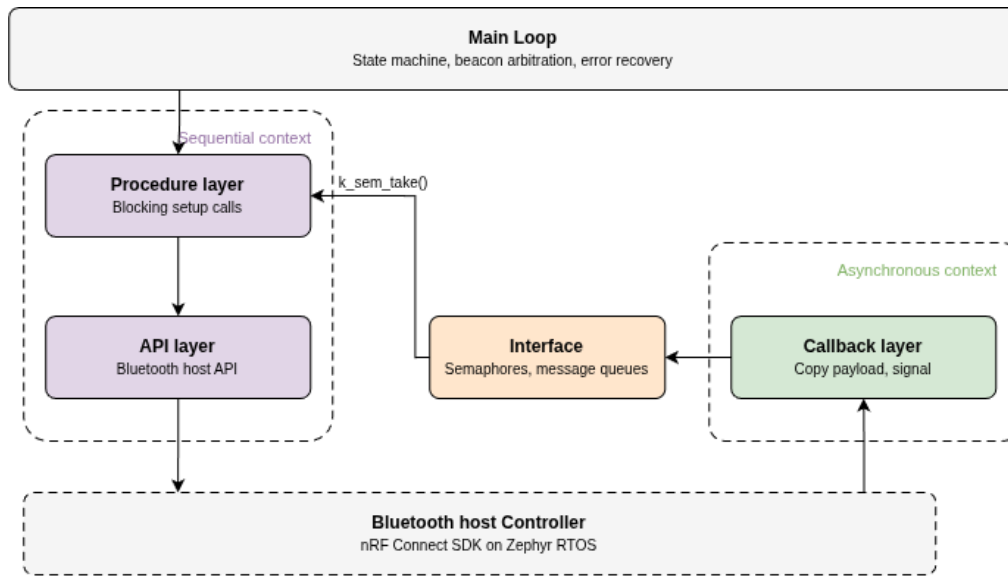


Figure 3: Software architecture of the embedded firmware: the API, Procedure and Callback layers bridge the sequential setup context and the asynchronous controller-callback context, orchestrated by the main state machine.

5.2 Configuration strategy

As seen in section 3.2: Anatomy of a CS procedure, with the architecture of a CS procedure we can establish the configuration strategy. And the first point to take care of is the ACL connection interval. This ACL has one main goal: wake up the link between two devices to keep the connection between specified intervals. This is a key function for the controller, and we cannot ignore it, but the issue is that if an ACL needs to be started, the controller will cancel the CS subevent to keep the link resulting in multiple errors and inefficiency. For this, we must estimate the time of each procedure to get the ACL outside of the subevents. The parameter setting up the timing is in the “.conf” file and is named: CONFIG_BT_CTLR_SDC_CENTRAL_ACL_EVENT_SPACING_DEFAULT.

The second important point is the procedure size. Two parameters set the bounds:

- Max_procedure_len: sets the maximum time that the controller allows for the procedure, each unit corresponds to 0.625ms. The spacing between two procedures is expressed by « min_procedure_interval » and « max_procedure_interval » which will be set at the same exact number to get deterministic time for all the procedures.
- Max_procedure_count: limits the number of procedures the controller executes before stopping; setting it to zero places the controller in a free running mode, in which procedures repeat indefinitely at the negotiated interval without any further intervention from the application.

The subtlety of max_procedure_len is that it acts like a hard ceiling. When the time allowed to the procedure is over, the procedure will stop and the controller will leave the remaining channels unmeasured. The procedure will do her report and complete itself. The less channel you do in the procedure the more uncertain and ambiguous the range will be, especially if you are doing only mode 2 (CS) without sub mode 1 (RTT). But this leads to a concept to reduce the length of the ranging procedures, which is channel mapping.

5.3 Channel map, ambiguity, resolution and precision

The channel map controls three main concepts, ambiguity, resolution, and precision, which allows the configuration to be optimized for each use case. In our firmware, the map is generated from a parameter called Channel Thinning, which builds the mask for the number of channels we want to keep. Two reductions are conceivable:

- Decimation keeps one channel out of N across the whole band: the full map has a 1 MHz spacing and a 150 m ambiguity, while a thinning factor of 3 gives a 3 MHz spacing and a 50 m ambiguity, the span and therefore the resolution being unchanged.
- Truncation instead keeps a contiguous block, which preserves the 150 m ambiguity but narrows the span and degrades the resolution.

It is important to know that the ambiguity follows $c/2\Delta f$ and the resolution $c/2 \cdot \text{span}$, while the precision follows the number of channels sampled, whichever reduction is used.

5.4 Application profiles

Parameters retained for the three use cases:

Parameters	Fast outdoor drone	Boat docking	Indoor Drone
CS mode	Main mode 2 submode 1	Mode 2 no submode	Mode 2 no submode
Channel thinning	3 (24 channels)	1,2 if n beacon ≥ 4	1,2 if n beacon ≥ 4
Mode 0 steps	2	3	3
Max main mode steps	6	10	10
Subevent length	4500us	8000us	8000us
Max procedure length	40 (25ms)	100 (62.5ms)	100 (62.5ms)
Channel map repetition	1	2 if n > 2 else 1	1
RTT type	32 bit random payload	No	No
Max tx power	Max	0	Max

Table 1: Channel Sounding parameters for the three application profiles.

5.4.1 Fast outdoor drone

This profile trades sampling density for airtime. One channel out of three is retained, which shortens the procedure by approximately the same factor and keeps the procedure window at 25 ms, so that the controller finds a placement easily even when several links compete for the timeline. The resulting 50 m alias period is the reason this profile is the only one that retains a sub mode: the interleaved mode-1 steps provide the RTT that selects the correct alias interval. The RTT to a 32-bit random payload increases accuracy by increasing the energy and the selectivity of the correlation. The RTT on a one beacon configuration less than one percent on the sampling rate dropping from 16.47Hz to 16.67 Hz.

5.4.2 Boat docking

The docking profile follows the opposite trade-off. The target is slow and close, so the map is left dense, the subevent is nearly twice as long, and three mode-0 steps are used instead of two to improve the estimation of the frequency offset. The alias period of phase based ranging alone then exceeds the docking envelope by a wide margin, so no RTT step is required, and the configuration uses mode-2 with no sub

mode. The transmit power is deliberately reduced to 0 dBm, since at a distance of a few meters the receiver would otherwise saturate and distort the phase measurement.

5.4.3 Indoor drone

The indoor profile shares the timing of the docking profile but retains the maximum transmit power, because the link budget in non line of sight conditions is the limiting factor. Frequency diversity is the primary rampart against multipath, which is a second reason to keep the dense map.

5.5 Scaling with the number of reflectors

The reflectors share the antenna time for the CS procedures, and the controller needs to place the ACL events to keep each link up. The spacing between the anchor points of two different connections is set by `CONFIG_BT_CTLR_SDC_CENTRAL_ACL_EVENT_SPACING_DEFAULT`. This spacing prevents collisions in the radio scheduling, since each cell has to hold one ACL event, one CS event and a guard interval, which for example gives 15 ms per link in the drone profile and 20 ms per link in the precision profiles. For two reflectors or more, the connection interval is therefore not a free parameter: it is the number of reflectors multiplied by the slot. The cell corresponds to the time slot the round-robin schedule reserves for one link. It must be large enough to hold that link's ACL event, its CS event, and a guard interval. The connection interval imposed on every link is the number of reflectors multiplied by the cell.

Concretely, with three reflectors sharing the drone profile's 15 ms cell, the controller lays out one 15 ms slot per link back to back: reflector 1's ACL event, CS event and guard, then reflector 2's, then reflector 3's, before returning to reflector 1. The full round therefore takes $3 \times 15 \text{ ms} = 45 \text{ ms}$, the connection interval enforced on every link as we can see in figure 4.

Scheduling for three reflectors (Fast outdoor drone profile)

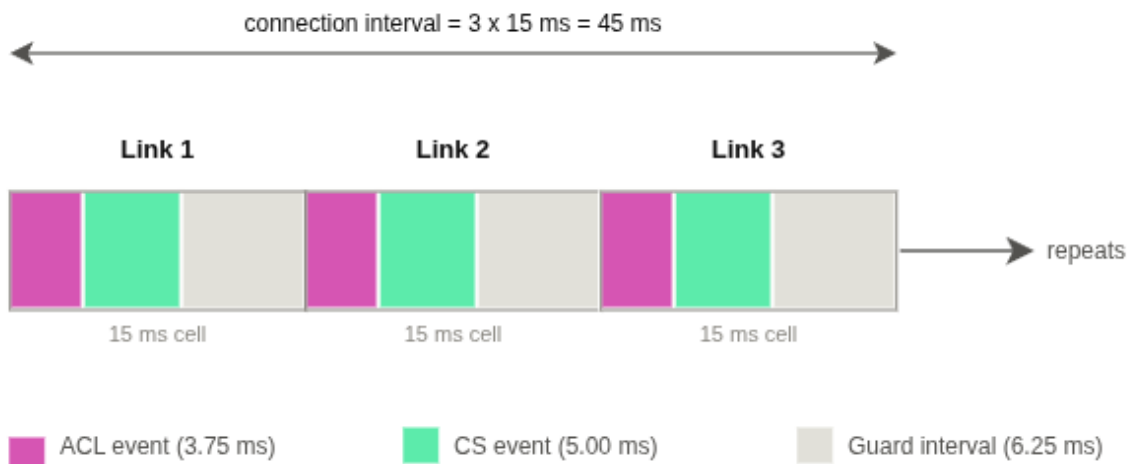


Figure 4: Scheduling three reflectors sharing the antenna time (Fast outdoor drone profile). Each 15 ms cell reserves an ACL event, a CS event and a guard interval; the connection interval is the cell multiplied by the number of reflectors.

The single reflector case is governed by a different constraint and is not a particular case of this rule. With one central connection there is no spacing to enforce between anchors, so the lower bound on the interval comes only from the requirement that one ACL event and one CS event fit inside it.

5.6 Controller reservations and the implicit effect of enabling CS

The application parameters are only one half of the configuration. The other half consists of the reservations that the SoftDevice Controller makes in its own timeline. These reservations are not merely limits they are the quantities the scheduler uses when it decides where to place an event, so an oversized reservation is as harmful as an undersized one.

An initially unnoticed behaviour is that enabling Channel Sounding changes one of these defaults implicitly. When the feature is enabled, the controller raises the default spacing between the ACL anchors of central connections to 90000 us, so that a CS event can always be placed behind each ACL event. This default is conservative and generic: it assumes a long CS event and a relaxed connection interval. It was not consistent with the target of this work, where the CS event is short, and the connection interval must remain small enough to sustain a usable refresh rate. With two links and a connection interval of 60 ms a 90 ms grid cannot be honoured, so the controller delayed the first subevent while searching for a free slot and rejected short connection intervals. The observed consequence was a setup latency of approximately 700 ms during which the radio was idle, which no change of the CS procedure parameters could reduce, because the cause lay in the scheduling reservations and not in the procedure itself.

Table 2 lists the six symbols that were set explicitly. The first three are timing reservations, which the scheduler uses directly when placing an event. The last three are memory allocations, one per link, which do not affect the placement but limit how many links can be armed at the same time. Those symbols are compilation time defined and cannot be hot swapped.

Kconfig symbol	Value	Role
CONFIG_BT_CTLR_SDC_CS_EVENT_LEN_DEFAULT	5000 us (8500 us with the precision overlay)	Reservation for one CS event, must cover max_subevent_len
CONFIG_BT_CTLR_SDC_CENTRAL_ACL_EVENT_SPACING_DEFAULT	15000 us (20000 us with the precision overlay)	Spacing between two ACL anchors of central links, implicitly 90000 us when CS is enabled but overridable
CONFIG_BT_CTLR_SDC_MAX_CONN_EVENT_LEN_DEFAULT	3750 us	Reservation for one ACL event per link, event extension preserves GATT throughput
CONFIG_BT_CTLR_SDC_CS_COUNT	5	Number of concurrent CS contexts, one per link
CONFIG_BT_CHANNEL_SOUNDING_REASSEMBLY_BUFFER_CNT	5	One host reassembly buffer per active link
CONFIG_BT_MAX_PAIRED	5	One session key slot per encrypted link, required even without bonding

Table 2: Controller reservations set explicitly in *prj.conf*.

After this resizing, the connection interval could be reduced from 60 ms to 30 ms for two links, and the setup latency fell from approximately 700 ms to approximately 340 ms with no aborted subevent. An attempt with a grid pitch of 10000 us and a connection interval of 20 ms failed: a guard of 1250 us per cell is insufficient once ACL retransmissions are considered and produced systematic aborts with reason code 0x3 on every link.

5.7 Concurrent free running procedures

The measurement pipeline rests on two mechanisms that were absent from our first implementation. The first is the free running mode of the procedure, obtained by setting the “max_procedure_count” to zero: the controller then repeats the procedure at every procedure interval until an explicit disable, so the link layer start procedure is paid once when the link is armed and never again per measurement. The second is

the allocation of one CS context per connection, which allows several procedures to run concurrently on distinct links.

Combined, they change the nature of the measurement loop. Every reflector holds a procedure permanently, so the setup cost is paid once per link instead of once per measurement. As each CS subevent finishes, the Controller reports its results straight to the Host, instead of waiting for the full procedure to complete. What was a sequence of separate measurements becoming a continuous stream, at the rate reported in Section 7.

6. Distance estimation from the raw IQ data

The embedded code does not return numerical values from the CS sequence. The initiator board returns an IQ sample of every tone of the sequence which is directly forwarded to the UART. The data transferred over UART can be processed through different methods and algorithms to get the best estimation of the distance.

The following section will explore a method to process those IQ measurements. This part is explained for common knowledge and to know how to process the data after the output. You have to keep in mind that the studies will not provide any form of numerical distance estimation.

6.1 What the IQ data contains

A CS procedure returns the sequence of steps presented in section 3.2.3: What constitutes a procedure. The mode 2 steps are carrying the phase-based ranging payload and will be used for the estimation. Each mode-2 tone is sounded on one CS channel k , at carrier frequency $f_k = 2402 + k$ MHz. At that frequency both devices measure a Phase Correction Term (PCT): a complex sample whose real and imaginary parts, each coded on 12 signed bits, encode the phase and the amplitude of the tone. The measurement is bidirectional, so the initiator measures the tone transmitted by the reflector, and the reflector measures the tone transmitted by the initiator.

Taken alone, neither of the two measurements is usable because the device generate its tone using its own physical quartz oscillator which are not sync on the same oscillation phase resulting in different phase reference of the signal, so each measurement carries the propagation phase plus an unknown phase offset between the two devices. That offset enters the two directions with opposite signs, since the roles of transmitter and receiver are exchanged. Summing the two phases therefore cancels it and leaves the round-trip phase, which is why the estimator works on the product of the two samples. The cancellation would be exact if the offset were the same at both measurement instants. Moreover, the two oscillators do not run at the same frequency, so it drifts slightly between the two exchanges, which is what the mode-0 steps measure and correct.

The complete input to one distance estimate corresponds to the set of sounded channels, the two complex samples and the quality flag of each tone, and if the sub mode 1 is selected, their timestamps are included and the mode-0 offsets and RSSI.

6.2 From phase to distance

The phase of a received tone is proportional to the propagation distance, as we saw on section 5.3: Channel map, ambiguity, resolution and precision there is a relation between the channel mapping with the ambiguity and the resolution. The phase is determined by the equation:

$$\varphi(f) = -(4\pi/c) \cdot f \cdot d$$

Here c corresponds to the speed of light, f the frequency of the wavelength and d the distance the factor 4π appears because the signal travels the distance twice. Inverting this relation gives the distance as the slope of the phase against frequency:

$$d = -(c/4\pi) \cdot d\varphi/df$$

Two ambiguities follow this, and they must not be confused. The first one comes from the carrier itself: the phase of a single tone is only modulo one rotation, so it repeats every half wavelength, about 6.15 cm at 2.44 GHz. A single tone therefore cannot tell which repetition it is in. This is precisely why Channel Sounding sounds many tones instead of one: the phase measured on a single frequency is ambiguous, but the rate at which it rotates from one frequency to the next is not, and that rate is what the equation converts into a distance.

The second one is the alias ambiguity introduced in section 5.3: because the tones form a comb of spacing Δf , the whole pattern repeats with period $c/2\Delta f$, that is 150 m with the dense map and 50 m with a thinning factor of 3.

6.3 The estimators

Now that the principle of phase-based ranging has been established, this section presents the two estimators used to convert the tone phases into a distance, with their respective advantages and limitations.

6.3.1 IFFT delay profile

The first estimator works in the delay domain rather than on the phase directly. The measured transfer terms are placed on the full channel grid, every channel that was not sounded being set to zero, and an inverse Fourier transform is applied. The result is a curve where the main peak is most likely the direct path. With this method we can estimate multipath with the number of peaks appearing after the IFFT. The distance is read from the position of the strongest peak.

Two properties make this estimator useful. It is immune to the carrier ambiguity of section 6.2, because it works on the envelope of the response rather than on the phase of any individual tone. And like said earlier, it makes the multipath visible: several peaks mean several paths, and a reflected peak taller than the direct one is the visible cause of a biased estimate, which a single distance value would have hidden.

Its limitation is its resolution. Two paths separated by less than the delay resolution merge into a single peak, and the position of that peak is not the position of either path. The estimate is therefore robust but coarse, which is why it is used as a starting point rather than as the final answer.

6.3.2 Bayesian phase likelihood

The second estimator asks a different question. Rather than transforming the measurements, it tests candidate distances: for a given candidate, it computes how likely the observed phases would be if the target were actually at that distance. The phase of each tone is treated as a noisy observation, and the contributions of all valid tones are combined into a single likelihood. The likelihood is high when the candidate makes the phases of every tone consistent with one another, and the distance that maximizes it is the estimate. Its precision is set by the sharpness of that maximum, of the order of a few centimeters.

The difficulty is that this likelihood is not unimodal. Because a phase repeats every half wavelength, the candidate distances that make the tones coherent form a comb of narrow lobes, all of them equally good

locally. The estimator therefore has to choose between them. Firstly, a local search starts with the IFFT value, which lands on the nearest lobe. Secondly, a global comparison of the lobe candidates over the unambiguous range, which keeps the most likely. This second step is what prevents the estimator from settling on a lobe that corresponds to a reflection rather than to the direct path.

The two estimators are therefore complementary: the IFFT is robust but coarse, the likelihood is precise but ambiguous, and the first is used to resolve the ambiguity of the second.

7. Tests and experimental results

7.1 Object and principle

The goal of the study is to evaluate the quality of the channel sounding tones, evaluating depending on the configuration the losses in terms of signal fidelity.

The choice to not conclude distances is deliberate. Estimating distances adds a lot of uncertainty due to the algorithms used, how they performed depending on the number of channels received, and other unknown variables.

To get higher fidelity of the results, we will measure the consistency of the IQ measurements. This allows us to observe if the IQ stays stable over time or drifts on a static setup. These results give us the limits for a future estimation algorithm who will suit our ranging method.

This makes the evaluation independent of any estimator and directly comparable across configurations.

Fidelity is not the only requirement. Our ranging method is meant to run on a moving platform, a drone or a boat, so the system must also keep a high enough sampling rate to follow the motion. This sets up a trade-off, because everything that improves the tone fidelity, such as adding channels, repeating the channel map or adding reflectors, also costs airtime and lowers the update rate. A usable configuration therefore has to balance the IQ consistency against the sampling frequency. For this reason, we report the per-link and the aggregate sampling rate together with the consistency, and we read the two side by side rather than optimizing the fidelity alone.

7.2 The metric: coherence R

For each channel k , over a static capture of T procedures:

$$R_k = | (1/T) * \sum_t \exp(j * \arg(Y_I(t) * Y_R(t))) | \in [0, 1]$$

- Y_I and Y_R are the complex IQ samples from both directions (initiator and reflector). The product $Y_I * Y_R$ is the two-way measurement: it cancels the common local oscillator offset.
- R_k is the mean resultant length (circular statistics), that is the coherence of the channel: $R = 1$ means a perfectly stable phase, $R = 0$ means a random phase. In practice, R is close to the quality of the tone.

The global score of a test is the median of the R_k . The 10th percentile points out the weakest (faded) channels.

One key property is that no calibration is needed. R only measures how stable the phase is over time, so any constant offset cancels out. The group delay and the absolute distance are both constant offsets, so they do not change R . The distance only matters as a test condition: it sets the link budget, hence the SNR, hence the fading, hence R .

7.3 Checking with the quantiles

The global metric does not tell us how many channels are usable. To see that, we sort the channels by decreasing coherence and plot the reliability vs kept-fraction curve (the coherence quantile):

X axis: % of channels kept, sorted from best to worst. Y axis: the worst R among that top-x %. How to read it: the point at $x = 20\%$ means "if we keep the 20 % most coherent channels, they all have $R \geq y$ ". The usual percentiles (10%, 20%, 50%, 80%) are marked on the curve.

What the curve checks:

- How many channels are reliable: a curve that stays high for a long time means many usable channels, a curve that drops fast means few.
- A ranking of the configurations: the area under the curve sums up the global reliability, and when several config curves are overlaid, they rank without ambiguity.

An important note: keeping fewer channels raises the average R but destroys the frequency diversity, which leads to lower precision and greater uncertainty. So the curve does not tell us to "keep the top 10%". It quantifies a quality versus diversity trade-off and shows which channels to remove (the fading dips), rather than reducing the set to a few tones.

7.4 Experimental Setup

The device under test is the nRF54L15DK running our firmware on NCS v3.3.0. One board acts as the initiator, and up to three boards act as fixed reflectors. All distances were measured with a laser rangefinder and used only as metadata, never for calibration. The IQ dump was read over the VCOM data port at 921600 baud, and the coherence was computed offline with our analysis tool.

We tested two of the three delivered profiles, each one at $N = 1$ and $N = 3$ reflectors. The profiles differ mainly by their channel map, which is what sets the frequency diversity:

Profile	Channel map	Notable feature
DRONE_FAST_OUTDOOR	thinning 3, about 24 channels	PBR plus RTT, short subevent
DRONE_INDOOR	full map, 72 channels	PBR only, max TX
BOAT_DOCKING	72 channels at $N = 1$, thinning 2 (36 channels) at $N = 3$	PBR only, channel map repetition 2 at N up to 2

Table 3: Channel map and notable configuration feature of the three tested profiles, used for the experimental setup.

Two environments were used: an open outdoor space in line of sight, and an indoor corridor. The corridor behaves like a waveguide with dense multipath, which is the stress case for frequency selective fading. Each capture lasted 30 s at $N = 1$ and 60 s at $N = 3$, because at $N = 3$ each channel is sampled in only a fraction of the procedures and needs more time to gather enough samples.

At $N = 3$ the three reflectors were placed at 5, 10 and 15 m at the same time, so a single acquisition gives the coherence of each link at three different link budgets, plus the multi-beacon effect (cross-link scheduling and partial per-procedure coverage). The metric requires a static channel, so any capture

disturbed by a moving object was rejected and repeated. Two captures were discarded for this reason, one when a vehicle passed outdoors and one when a bird approached a reflector.

7.5 Results

The tables below give, for each condition, the median and the 10th percentile of R , together with the RSSI as the link condition. For the $N = 3$ tests, the per-link distance is recovered from the RSSI order (the $b_0/b_1/b_2$ index depends on the connection order, not on the position).

7.5.1 DRONE_FAST_OUTDOOR

N	Distance	RSSI (dBm)	R median	R 10th
1	5 m	-63.9	0.985	0.981
1	10 m	-79.2	0.896	0.736
3	5 m	-52.4	0.993	0.992
3	10 m	-79.3	0.944	0.814
3	15 m	-84.1	0.515	0.172

Table 4: Coherence results for the DRONE_FAST_OUTDOOR profile at $N = 1$ and $N = 3$ reflectors.

7.5.2 DRONE_INDOOR

N	Distance	RSSI (dBm)	R median	R 10th
1	5 m	-58.7	0.970	0.955
1	10 m	-75.7	0.807	0.468
3	5 m	-59.2	0.976	0.968
3	10 m	-64.1	0.976	0.878
3	15 m	-74.0	0.878	0.665

Table 5: Coherence results for the DRONE_INDOOR profile at $N = 1$ and $N = 3$ reflectors.

7.5.3 BOATDOCKING_INDOOR

N	Distance	RSSI (dBm)	R median	R 10th
1	5 m	-55.5	0.970	0.965
1	10 m	-71.7	0.954	0.834
3	5 m	-57.6	0.951	0.925
3	10 m	-66.7	0.876	0.376
3	15 m	-73.6	0.822	0.451

Table 6: Coherence results for the BOATDOCKING_INDOOR profile at $N = 1$ and $N = 3$ reflectors.

7.5.4 BOATDOCKING_OUTDOOR

N	Distance	RSSI (dBm)	R median	R 10th
1	5 m (geometry B)	-52.5	0.994	0.993
1	15 m	-79.4	0.926	0.601
3	5 m	-52.5	0.994	0.992
3	10 m	-78.7	0.942	0.489
3	15 m	-83.5	0.634	0.390

Table 7: Coherence results for the BOAT DOCKING profile at $N = 1$ and $N = 3$ reflectors.

7.5.5 Measured sampling rate

Profile	Per-beacon rate at $N = 1$	Per-beacon rate at $N = 3$	Aggregate rate at $N = 3$ (3 links)
DRONE_FAST_OUTDOOR	16.5 Hz	5.0 Hz	15.0 Hz
DRONE_INDOOR	11.1 Hz	3.9 Hz	11.7 Hz
BOAT DOCKING	6.2 Hz	3.9 Hz	11.7 Hz

Table 8: Measured per-beacon and aggregate sampling rate for each profile at $N = 1$ and $N = 3$.

7.6 Observations

7.6.1 Coherence is driven first by the SNR

Across every condition, R follows the RSSI. On a strong link at 5 meters and with a RSSI around -52 to -59 dBm the coherence saturates above 0.97 whatever the profile and the environment. On the weakest link of the tests, we have a R median of 0.515 and a 10th percentile at 0.172. The figure 5 plot R against the per-channel amplitude for all tests at once, both environments fall on a single R versus amplitude trend. The practical consequence is that at 5 m the profiles cannot be told apart, since the coherence is capped by the SNR. The configuration differences only show up at range, on a weak link.

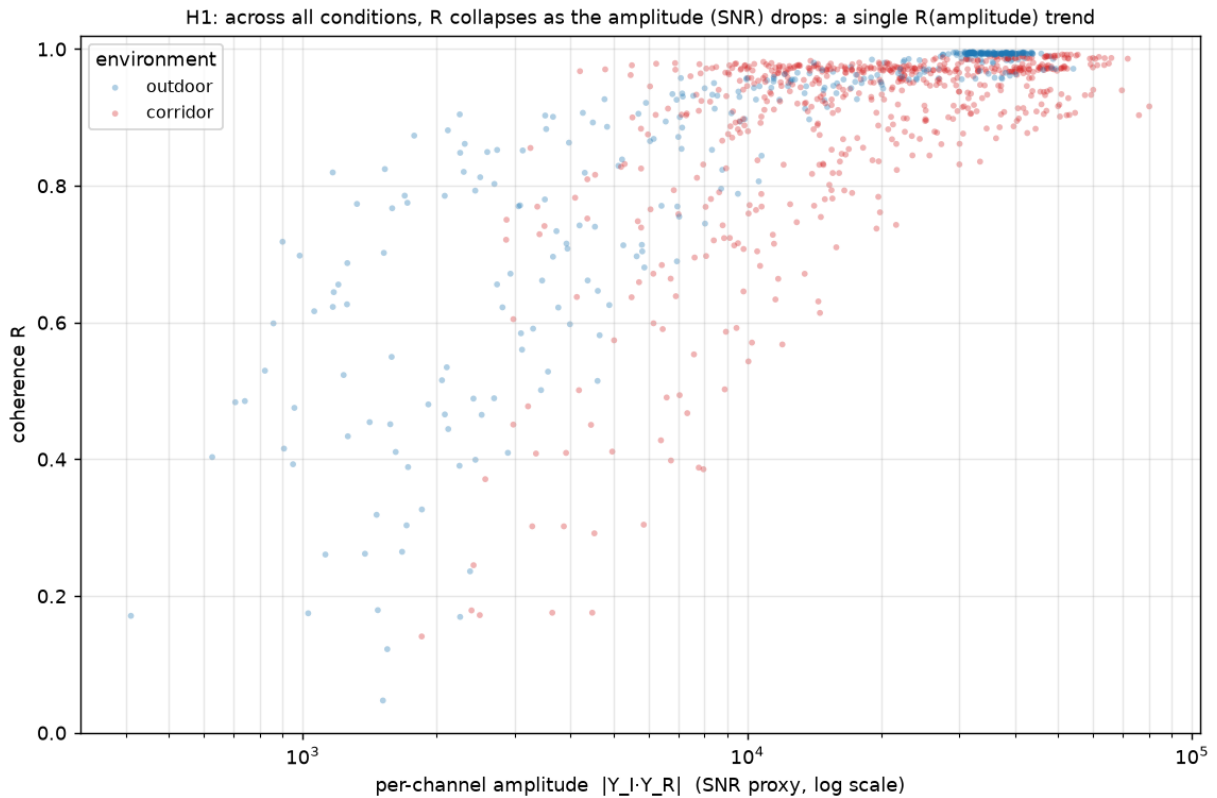


Figure 5: Coherence R versus per-channel amplitude (SNR proxy) across all outdoor and corridor tests, showing a single $R(\text{amplitude})$ trend regardless of profile or environment (H1).

7.6.2 Multipath lowers the coherence at equal SNR

The clearest way to see this is to compare the same profile in the two environments. For the boat profile at $N = 1$ and 15 m, the corridor has a better link budget than the outdoor space (-71.7 against -79.4 dBm, an effect of the waveguide), yet its coherence is lower and its 10th percentile drops much more as we can see in figure 6. So it is not the SNR that degrades R here, it is frequency selective fading: some channels fall into the nulls of the multipath. The full 72-channel map reveals them, the median stays acceptable, but the low tail betrays the multipath. Figure 7 shows the per-channel view, a strong link keeps R flat near 1 while the amplitude varies, whereas a faded link shows R dipping exactly at the amplitude nulls.

H2: multipath at comparable SNR: the corridor has a better RSSI but a lower worst case (faded channels)

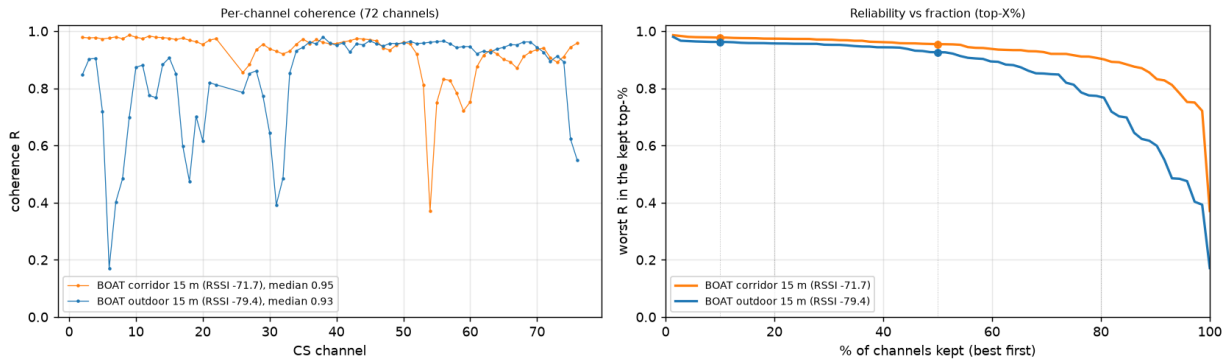


Figure 6: Multipath at comparable SNR: per-channel coherence and reliability-vs-kept-fraction for the boat-docking link, corridor versus outdoors (H2).

Per-channel coherence: strong link (flat, near 1) vs faded link (R dips at the multipath nulls)

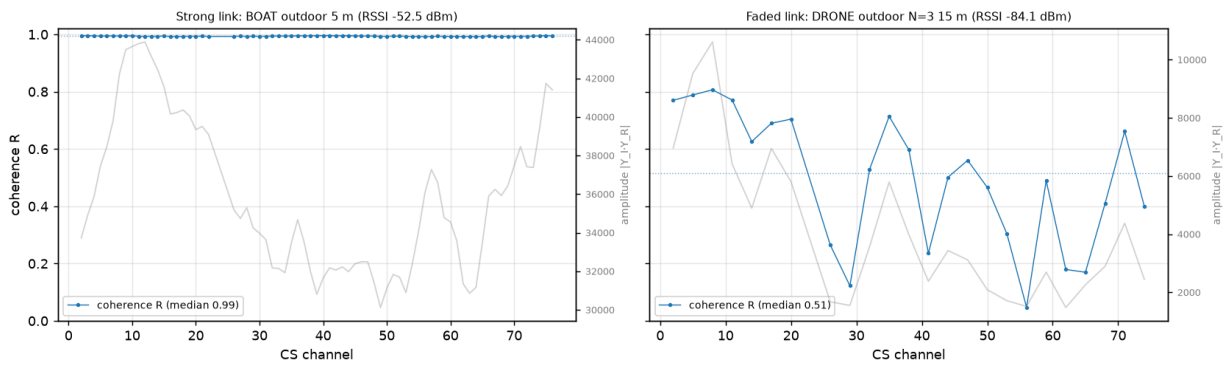


Figure 7: Per-channel coherence for a strong link (flat, near 1) versus a faded link (R dips at the multipath nulls).

7.6.3 The multi-beacon mode does not degrade the per-link consistency, and the sampling rate

At a fixed geometry, the coherence of each link at $N = 3$ matches the one measured at $N = 1$ as showed in figure 8.

Single vs multi-beacon: the per-link coherence at $N=3$ matches $N=1$ (same channel map)

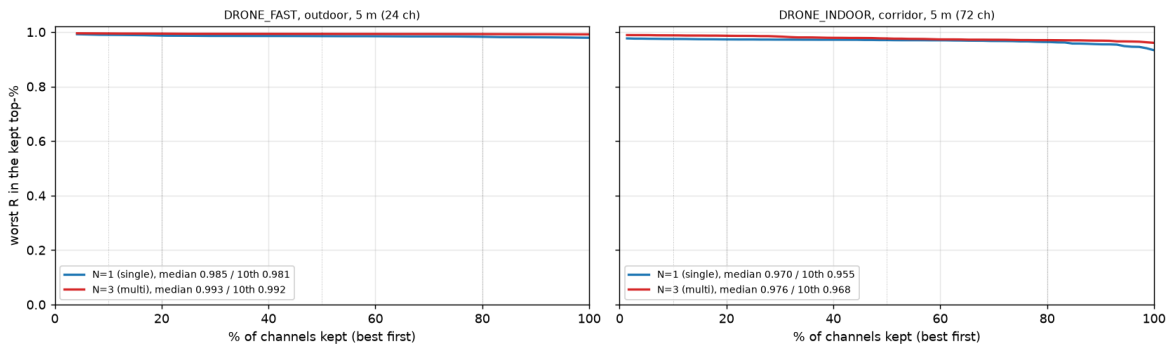


Figure 8: Per-link coherence at $N=3$ matches $N=1$

Going to $N = 3$ lowers the per-beacon rate, since the connection interval is multiplied by N , and each reflector is served once per round. For the drone and indoor profiles, whose channel map does not change with N , the per-beacon rate scales close to $1/N$ (16.5 Hz drops to 5.0 Hz, 11.1 Hz drops to 3.9 Hz).

The important point is that the aggregate sampling rate, summed over all the reflectors, is preserved. For the drone profile the total goes from 16.5 Hz at $N = 1$ to about 15.0 Hz at $N = 3$, so we measure three links for essentially the same total airtime as a single one. In other words, the multi-beacon count is a rate factor per link, not a consistency factor, and it parallelizes the acquisition at no cost in total.

The boat profile is a special case that even gains aggregate rate: it goes from 6.2 Hz at $N = 1$ to 11.7 Hz aggregate at $N = 3$. This is not a free lunch but a consequence of the profile itself changing with N . At $N = 1$ the boat map keeps the full 72 channels with a channel map repetition of 2, which makes each procedure long and the rate low. At $N = 3$ the repetition drops to 1 and the map is thinned to 36 channels, so each procedure is much shorter and the per-beacon rate stays near 3.9 Hz instead of falling to $6.2/3$. The gain in aggregate rate is therefore paid by a lighter per-procedure map, which is the same trade-off seen on the coherence side (density versus worst case).

7.6.4 The real configuration lever is the channel resampling

Two opposite mechanisms appear. First, channel map repetition raises the median at low N . At $N = 1$ the boat profile measures each channel twice per round and keeps 72 channels, which places it well above the indoor profile at 15 m (0.954 against 0.807 on the median, 0.834 against 0.468 on the 10th percentile). Part of the gap comes from a 4 dB RSSI advantage, but not all of it, since moving the 10th percentile from 0.468 to 0.834 is consistent with an averaging effect that smooths the per-channel phase noise. Second, thinning arbitrates the worst case. At $N = 3$ the boat profile loses both assets (the repetition drops to 1 and the map is thinned to 36 channels), and the denser indoor map moves ahead. Compared to the drone profile outdoors at $N = 3$ with matched RSSI, the sparse 24-channel drone map gives a much better 10th percentile at mid range (0.814 against 0.489 at 10 m), because it avoids the fading dips, while the denser map captures more channels including faded ones that pull the low tail down (Figure 7). This is the density versus worst case trade-off: more channels means more frequency diversity, useful for resolution and for the ambiguity period, but also more dead channels that weigh on the low tail.

7.7 Limitations

The outdoor boat capture at 5 m was taken with a different geometry (different angle and position), which gave an RSSI of -52.5 dBm against -63.9 dBm for the drone at 5 m, which results in an 11 dB gap at the same distance. The direct boat against drone comparison at 5 m outdoors is therefore not clean, and it mostly illustrates that distance alone does not set the SNR. Each condition was measured once with a 30 or 60 second procedure recording, so we do not have error bars and cannot state the repeatability of R .

7.8 Conclusion

The study establishes the physical drivers of the tone quality. The coherence is governed first by the SNR, then degraded by frequency selective fading whenever multipath is present, and this degradation shows up in the low tail of the coherence rather than in the median. On the acquisition side, the multi-beacon mode preserves the per-link consistency and keeps the aggregate sampling rate close to the single-beacon case, so three reflectors are obtained at essentially no cost in total throughput. The main configuration lever is the way each channel is resampled, through the channel map repetition and through the density of the channel map.

These findings must be read against a second requirement. Trying to maximize consistency may not be the end in itself. On drone application, the initiator is moving and needs a high enough update rate to follow the motion. Every setting that raises the fidelity, such as adding channels, repeating the channel map or adding reflectors, also costs airtime and lowers the rate. Choosing a configuration is therefore a trade-off between the IQ fidelity and the sampling frequency, not a search for the highest coherence alone. A slow but very consistent acquisition would be useless on a drone, and a fast but noisy one would not feed a reliable estimator. A system that could fusion the data from the CS procedures and equipped with IMU could be complementary to the CS procedure may correct the natural IMU drift.

This trade-off is exactly what an adaptive channel map addresses. The low 10th percentile reveals the faded channels, which carry little information yet still cost airtime. Removing them raises the fidelity and frees time that can be spent either on a higher update rate or on the channels that are still reliable, while keeping enough channels to preserve the frequency diversity that the resolution and the ambiguity period require. The map should therefore be trimmed at the fading dips, not reduced to a handful of tones.

The results above give the limits that a future estimation algorithm will have to work within, and they point to configurations that balance fidelity and rate rather than maximizing either one.

8. Conclusion

This internship set out to answer a question: how can the Channel Sounding feature introduced in BLE Specification 6.0 be configured and implemented to deliver fast, reliable raw ranging data for a ground-to-drone link, within the constraints a real embedded controller imposes on multi-beacon scheduling? The answer developed across this report is a firmware architecture built on three tuned application profiles (fast outdoor drone, boat docking, indoor drone), each trading channel density, subevent length and transmit power against the operational constraints of its use case, this combined with a free-running CS procedure we managed to get continuous stream of IQ measurements.

The experimental results in section 7 validate the physical reasoning behind this design rather than a final distance-estimation pipeline, which was explicitly out of scope. Coherence R, measured directly on the raw IQ data with no calibration required, confirmed that signal quality is governed first by SNR and only then degraded by frequency-selective fading when multipath is present, and that this degradation shows up in the low tail of the coherence distribution rather than in the median. It also confirmed that the multi-beacon mode does not degrade per-link consistency, which is one of the most important learning of these studies.

The research work shows that BLE even if it is a technology never designed for ranging, it can be configured for IQ sampling and leading to a simple way to range precisely distances between two BLE systems.

This internship gave me the opportunity to dip into the world of radiofrequencies. Working through the Channel Sounding specification taught me how to find the levers a real system exposes and gave me the knowledge to design a coherent firmware architecture that optimizes CS sequences within the constraints of an actual radio controller. Along the way I also learned to work with Zephyr RTOS, which is a powerful full OS to dip into embedded application. Beyond the technical skills, this project taught me how to read and apply a large, unfamiliar specification directly to hardware, and how to turn ambiguous requirements (accuracy, update rate, multi-beacon scaling) into a set of concrete, testable configuration choices.

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